

WhatsApp Style Customer Chat Platform - Software Requirements Document (SRD)

Project Overview

Develop a complete customer communication platform similar to WhatsApp Business Chat where businesses can communicate with customers through a website chat widget. The platform should include live chat, lead management, CRM, automation, multilingual support, analytics, exports, and WhatsApp handoff while keeping customer data secure.

User Roles

1. Super Admin

- Full platform access
- Manage all businesses
- View system analytics
- Manage subscriptions
- Platform settings

2. Business Admin

- Access only their own company
- Manage agents
- View all company chats
- Manage automation
- Export data
- Broadcast messages
- CRM access

3. Agent

- Access only assigned conversations
- Reply to customers

- Update lead status
- Add notes
- Cannot access other company data

4. Customer

- Chat from website
 - Upload profile picture
 - Create display name
 - Continue previous conversations
 - Receive automated messages
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Customer Chat Features

User Identity

Instead of showing only anonymous users:

Customer should be able to:

- Enter Name
- Mobile Number
- Email (Optional)
- Company (Optional)
- Profile Picture Upload

Example:

 Rahul Sharma

Instead of:

Anonymous Visitor

Chat UI

WhatsApp-like interface

Features:

- Profile Picture
 - Online indicator
 - Typing indicator
 - Read receipts
 - Delivered status
 - Message timestamp
 - Emoji support
 - Image sharing
 - Document sharing
 - Video sharing
 - Audio sharing
 - Voice notes
 - Drag & Drop upload
 - Reply to message
 - Forward message
 - Search messages
 - Pin conversation
 - Archive conversation
 - Delete conversation
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Auto Welcome Message

If customer opens chat

Automatically send:

Hello 🖐️

Welcome to ABC Company.

How can we help you today?

Please choose an option:

1. Sales
2. Support
3. Billing
4. Talk to Human

No agent required.

Customer Details Collection

If user has not entered details

Ask:

- Name
- Mobile Number
- Email

Save automatically.

Never ask again.

Customer Identification

Customer should be recognized using:

- Mobile Number
- Email
- Browser Token
- Login ID (optional)

Whenever customer returns

Previous chat should load automatically.

Offline Handling

If no agent is online:

Display:

Currently our team is offline.

You can continue chatting or contact us on WhatsApp.

Buttons:

- Continue Chat
- WhatsApp Now

Clicking WhatsApp should open WhatsApp with predefined message.

Example:

Hi,
I visited your website and need support.

WhatsApp Redirect

If customer clicks WhatsApp

Automatically include:

Customer Name

Customer Phone

Visited Page

Lead ID

Current Issue

Time

Reference Link

Continue Conversation

When customer later returns

Old conversation should appear.

Nothing should be lost.

Multi Agent System

Every business can create multiple agents.

Example:

Sales Agent

Support Agent

Billing Agent

Manager

Each agent gets:

Own Login

Own Dashboard

Own Chats

Own Permissions

Data Isolation

Very Important

Business A

Cannot access

Business B data

Business B

Cannot access

Business A data

Every company data must be completely isolated.

Only Super Admin can access all businesses.

Chat Assignment

Chats can be

Auto Assigned

or

Manual Assigned

Rules:

Round Robin

Least Busy Agent

Department Wise

Manual

Lead Management

Every chat automatically becomes a Lead.

Lead fields:

Lead ID

Name

Phone

Email

Company

Source

Website Page

Campaign

UTM Source

UTM Medium

UTM Campaign

Country

City

Browser

Device

OS

Language

Date

Last Activity

Status

Tags

Assigned Agent

Priority

Notes

Lead Pipeline

Status:

New

Contacted

Interested

Qualified

Proposal Sent

Won

Lost

Spam

Follow Up

Custom Status

CRM

Complete CRM

Customer Profile

Timeline

Previous Chats

Orders

Invoices

Notes

Tasks

Documents

Follow-ups

Internal Notes

Call Logs

Broadcast Messaging

Business Admin can send broadcast to:

All Customers

Filtered Customers

Tags

Lead Status

Country

Language

Date Range

Custom Segment

Support:

Text

Image

Video

PDF

Buttons

Templates

Schedule Broadcast

Automation Builder

Visual Flow Builder

Example:

Customer opens chat



Send Welcome Message



Ask Language



If English



English Flow



If Hindi



Hindi Flow



Collect Phone



Assign Agent



Notify Team



Create Lead



Send Email



Send WhatsApp



End

Automation Features

Trigger:

First Visit

Returning Customer

Lead Created

Lead Updated

No Reply

Agent Assigned

Order Created

Payment Received

Custom Event

Actions:

Send Message

Assign Agent

Create Lead

Send Email

Send WhatsApp

Webhook

Delay

Condition

Tag User

Create Task

AI Features

Automatic Language Detection

Hindi ↔ English Translation

Urdu Support

Gujarati

Punjabi

Marathi

Tamil

Telugu

Bengali

Auto Translate Incoming Messages

Auto Translate Outgoing Messages

Agent sees selected language.

Customer sees original language.

Smart AI

Suggested Replies

Auto Summary

Conversation Summary

Intent Detection

Sentiment Analysis

Spam Detection

Priority Detection

Knowledge Base Search

AI Reply Assistant

Share Chat Link

Every customer gets unique Chat ID.

Example:

chat.company.com/c/ABC12345

When shared

Open same conversation

Customer continues previous chat.

Website Landing

If user clicks shared link

Landing page opens

Shows:

Company

Previous Chat

Continue Chat

WhatsApp Button

Support Button

Call Button

Email Button

Contact Buttons

Support Button

Open Chat

WhatsApp Button

Open WhatsApp

Call Button

Call Business

Email Button

Compose Email

Customer Analytics

Track:

First Visit

Last Visit

Session Time

Total Visits

Pages Visited

Current Page

Country

City

Timezone

Browser

Device

Operating System

Referral Source

Campaign

UTM Parameters

Click Events

Downloads

Forms Submitted

Chat Duration

Response Time

Resolution Time

Conversion

Revenue (if integrated)

Dashboard

Charts:

Today's Chats

Active Chats

Waiting Chats

Missed Chats

New Leads

Qualified Leads

Revenue

Conversion Rate

Agent Performance

Response Time

CSAT

Languages

Traffic Sources

Top Pages

Export

Export:

CSV

Excel

PDF

JSON

XML

Filters:

Date

Agent

Lead Status

Country

Campaign

Tags

Department

Search

Global Search

Search by:

Name

Phone

Email

Lead ID

Message

Company

Tags

Agent

Notifications

Browser Notification

Desktop Notification

Email Notification

WhatsApp Notification

Push Notification

Sound Alerts

Security

JWT Authentication

Role Based Access Control (RBAC)

Two Factor Authentication (2FA)

Password Encryption

HTTPS

Audit Logs

IP Restrictions

Session Management

Data Encryption at Rest

APIs

REST API

Webhook Support

GraphQL (Optional)

Public API

Private API

Integrations

WhatsApp Business API

Email (SMTP)

Google Sheets

Zapier

Slack

Telegram

Razorpay

Stripe

Meta Pixel

Google Analytics

Google Tag Manager

Webhooks

Performance Requirements

- Support 100,000+ customers.
 - Real-time messaging with WebSockets.
 - Fast message delivery (<1 second where possible).
 - Responsive UI for desktop and mobile.
 - Scalable multi-tenant architecture.
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Technology Recommendation

Frontend:

- React / Next.js
- Tailwind CSS
- TypeScript

Backend:

- Node.js (NestJS or Express)
- Socket.IO / WebSockets

Database:

- PostgreSQL
- Redis (cache & queues)

Storage:

- AWS S3 or Cloudflare R2

Search:

- Elasticsearch or Meilisearch

Authentication:

- JWT + Refresh Tokens

Deployment:

- Docker
 - Kubernetes (optional)
 - Nginx
 - CI/CD pipeline
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Future Features

- Voice Calling
- Video Calling
- Screen Sharing

- AI Voice Bot
- AI Calling Agent
- Knowledge Base
- Ticketing System
- Mobile Apps (Android & iOS)
- White-label Branding
- Custom Domains
- Multi-brand Support
- Subscription & Billing Module
- Marketplace for Integrations
- Advanced Workflow Builder